ABHAY KUMAR SINGH

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PROFESSIONAL SUMMARY

Full Stack Web Developer with hands-on experience building scalable, production-ready web applications using React.js, Next.js, Node.js, and cloud-native microservices architecture. Proficient in REST API development, real-time communication (WebRTC, Socket.io), event-driven systems (Apache Kafka, Redis), and containerized deployments (Docker, Kubernetes). Passionate about delivering high-performance, user-centric software with clean code and modern DevOps practices.

TECHNICAL SKILLS

- **Programming Languages:** JavaScript (ES6+), TypeScript, Java, Python, C, SQL
- **Frontend Development:** React.js (Vite), Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design, WebRTC
- **Backend Development:** Node.js, Express.js, REST API Design, GraphQL basics, Socket.io, Apache Kafka, Redis (Caching/Pub-Sub)
- **Databases:** MongoDB, PostgreSQL, MySQL, Appwrite, Neon DB, Database Design, Query Optimization
- **DevOps & Cloud:** Docker, Kubernetes, AWS (EC2, S3, Cloud Foundations), CI/CD Pipelines, Linux/Bash
- **Developer Tools:** Git, GitHub, Postman, JWT Authentication, VS Code, npm/yarn
- **AI & Integrations:** OpenAI API, Python ML (Recommendations), LLM Integration
- **CS Fundamentals:** Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, System Design

PROJECTS

HireHeaven – Full Stack AI-Powered Job Portal | Microservices Architecture *Oct 2025– Feb 2026*

Tech Stack: Next.js, Node.js, PostgreSQL (Neon DB), Apache Kafka, Redis, Docker, Kubernetes, OpenAI API, Python ML

- Architected and developed a production-ready, scalable full stack job portal using Next.js (frontend) and event-driven microservices backend, handling high-concurrency workflows with Apache Kafka message queues and Redis caching layer.
- Designed and implemented RESTful APIs with Node.js/Express.js for job listings, user authentication (JWT), and application management; integrated PostgreSQL (Neon DB) with optimized queries for relational data.
- Built AI-powered Resume Analyzer feature using OpenAI API, providing ATS scoring, keyword optimization, and gap analysis to improve candidate match rates.
- Developed Intelligent Career Guide module delivering personalized learning roadmaps using LLM integration and Python ML algorithms for job recommendation engine.
- Containerized all microservices with Docker and orchestrated deployments using Kubernetes, ensuring zero-downtime rollouts and horizontal scalability.

Chattrix – Real-Time Full Stack Voice & Video Chat Application *May 2025 – Sep 2025*

Tech Stack: React.js, Node.js, Socket.io, WebRTC, Redis Pub/Sub, Docker, JWT, MongoDB

- Engineered a low-latency, peer-to-peer voice/video calling web application using WebRTC (Simple-Peer) for P2P media streaming with <200ms latency across global connections.
- Implemented real-time WebSocket signaling server with Socket.io and Redis Pub/Sub for scalable online presence tracking across Dockerized multi-server deployments.
- Developed secure JWT-based authentication system with persistent chat storage (MongoDB), comprehensive reconnection logic, and intuitive responsive UI built with React.js.
- Architected backend with Node.js/Express.js, ensuring fault-tolerant error handling and seamless media sharing capabilities with user presence indicators.

Blog Application – AI-Powered Full Stack Blog Platform *Dec 2024 – March 2025*

Tech Stack: React.js (Vite), Tailwind CSS, Appwrite, OpenAI API, GitHub OAuth

- Built a fast, mobile-first blog platform with React.js (Vite) frontend, leveraging Tailwind CSS for fully responsive, cross-device UI/UX with optimized Core Web Vitals.
- Integrated OpenAI API for automated AI content generation, reducing content creation time and enabling intelligent draft suggestions for blog authors.
- Implemented GitHub OAuth authentication with password reset workflow using Appwrite BaaS for database/storage management, ensuring secure user data handling and GDPR-compliant practices.

EDUCATION

BTech – Computer Science and Engineering | CGPA: 8.77/10 *Aug 2023 – May 2027*

Parul Institute of Technology, Parul University — Vadodara, Gujarat

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Software Engineering

CERTIFICATIONS & ACHIEVEMENTS

- AWS Academy Graduate – Cloud Foundations (Aug 2025, 20 hrs) — Amazon Web Services
- NPTEL – Computer Networks and Internet Protocol, Elite + Silver (79%) — IIT
- Artificial Intelligence Fundamentals – IBM SkillsBuild (NLP, Computer Vision, Machine Learning, Deep Learning)

KEY COMPETENCIES

Full Stack Development • REST API Design • Microservices Architecture • Real-Time Web Applications • Cloud Deployment • Agile/Scrum • Performance Optimization • System Design • CI/CD • Code Review • Cross-functional Collaboration